Synthetic Laws: Statistical Lawhood Under Finite Description

Gerard McCaul
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Theories idealise: any theory selects a finite set of variables, declares the rest invisible, and writes its laws in the compressed representation that remains. This paper derives the form those laws must take—and finds that consistency under changes of description uniquely forces the relative-entropy measure and the free-energy variational family, then systematically classifies the residue left when closure fails.

In a finite positive-support setting, four requirements on scalar law-content—relabelling invariance, atomwise locality, neutral refinement, and staged elimination—force the comparison between descriptions to the KL form; adding visible constraints forces the variational selector to the free-energy family. The visible generator $K_s = \sum_a g_a A_a$ is the cost function of retained complexity, pricing the distinctions the finite description keeps above the recursively generated null background. These are consequences of the structure of finite staged description, not modelling choices.

The *null law* at each scale is generated recursively by the same coarse-graining that defines the scale—the minimally committed baseline produced when lower-scale structure is absorbed without imposing additional visible constraints. When a fine law cannot be represented exactly at a coarser scale, the *closure residue* $\Delta_C = CG_\mu - G_M C$ composes coherently across scale towers. Its structured modes are renormalised couplings, memory kernels, noise, emergent order parameters, gauge freedom, holonomy, and spectral noncommutativity: a complete classification of effective physics as structured closure failure.

Taking the infinitesimal scale limit yields a residue density ω_s whose off-diagonal component drives spectral-frame rotation. The dimensionless ratio χ_s of off-diagonal residue to spectral gap measures distance from classical closure. When χ_s flows to a nontrivial fixed point, the failure of classical closure is itself scale-stable. Ordinary quantum mechanics is identified as the unique rigid case in which that fixed-point residue is scalar, central, and universal: the standard Hilbert-space formalism is the algebra of self-similar closure failure.

I. FINITE LAWHOOD AND RECURSIVE CLOSURE

Theories idealise. Any theory selects a finite set of variables, declares the rest invisible, and writes its laws in the compressed representation that remains. Formally this is a coarse-graining; every object of theory—particle trajectories, density matrices, continuum fields, block-spin configurations—is a compressed image of phenomena richer than itself.

Which kind of idealisation a theory performs depends on the relation between its register and what it describes (Fig. 1). Let Ω be the configuration space of the world, $R \subset \Omega$ the physical register carrying the theory, and $\pi : \Omega \to X$ the symbol map by which the register represents the world. Three regimes are structurally distinct: a *disjoint* theory has a register lying entirely outside its target; a *coupled* theory partially overlaps its target; a *reflexive* theory lies inside it, so that $R \subset \Omega$ and the target is Ω itself. Fundamental physics, cosmological theories, computational theories, and biological self-models all occupy the reflexive category.

A finite reflexive theory compresses: $|X| < |\Omega|$. Exact reflexive self-description would require the finite compressed image to contain its own representational action without residue—an exact self-similarity or fixed-point condition. Generically no such fixed point exists. But the very fact that a reflexive theory *can be written* is itself constraining: a consistent finite self-description must close under its own application, and wherever closure fails the failure is not arbitrary—it is structured residue whose composition laws determine whether the theory can remain predictive. The present paper asks what those laws must be.

The question this forces is not epistemological. It is structural:

> *How can a finite idealisation stand in lawful relation to phenomena richer than itself?*

Let X_μ denote the fine state space and X_M the coarser state space of the theory. A representation is a map

$$C : X_\mu \to X_M. \tag{1}$$

The fibre $C^{-1}(m)$ collects all fine states the theory declares equivalent. Whether the coarse description can generate its own dynamics—without constant reference to the finer level—is the *closure question.*

Ordinary symmetries act *within* a chosen representation: coordinate changes, basis rotations, canonical transformations. These are reversible relabellings that leave the partition C intact. But a theory also changes its own representation: coarse-graining, marginalisation, renormalisation, tracing out degrees of freedom. These are irreversible changes of description—the partition itself is replaced, and discarded information does not return. The question of lawhood is a question about the second kind.

A representation *closes exactly* when its variables contain enough information to generate their own evolution.

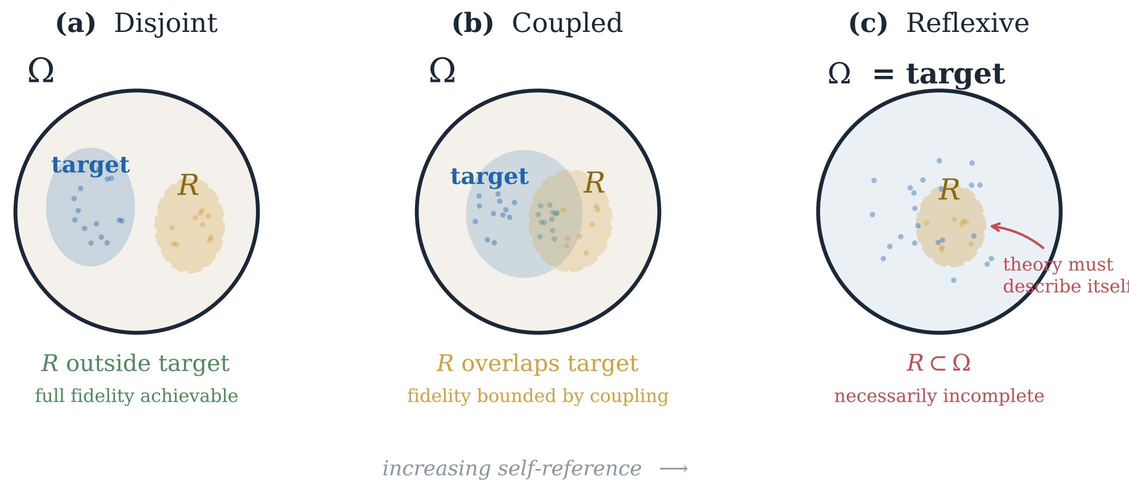

FIG. 1. **Three representational regimes.** *Disjoint:* the register R lies entirely outside its target; the theory and the modelled system share no state space. *Coupled:* R partially overlaps its target. *Reflexive:* $R \subset \Omega$ and the target is Ω itself. Fundamental physics, computation, and biological self-models are reflexive. A finite reflexive theory can close exactly only at a self-similar fixed point; generic finite compression leaves structured residue whose composition laws are the subject of this paper.

If G_μ is the fine generator and G_M the coarse generator, exact closure requires

$$CG_\mu = G_M C. \tag{2}$$

This says: evolving finely and then coarse-graining gives the same result as coarse-graining first and then evolving with the coarse law.

When (2) fails, define the *closure residue*

$$\Delta_C = CG_\mu - G_M C. \tag{3}$$

The closure residue is the determinate part of fine dynamics not internally generated by the coarse law, but still relevant for predictions. It is not a defect to be suppressed; it is the structured mismatch whose composition properties determine how finite descriptions can remain predictive across changes of representation.

A law is not merely a model written at one scale. It is a law-*form* that must be stable under changes of description: whatever variables survive at a coarser level, they must be selected by the same variational grammar as the fine variables. This is the higher-order scale principle the paper makes precise. The finite positive-support setting is not an incidental toy model: it is the exact setting of finite description. Continuum and field-theoretic theories are projective or limiting extensions of this finite calculus, not its prior foundation. The following sections ask, in sequence, what stability of the law-form forces.

II. STAGED ELIMINATION FORCES KL

Any distribution p^* over a finite state space is the unique minimiser of $J[\rho] = D_{\mathrm{KL}}(\rho\|p^*)$, which vanishes at $\rho = p^*$ and is positive elsewhere; constrained equilibrium distributions minimise a free-energy functional subject to the constraints. The variational formulation is therefore not a restriction on which states are physical—it is universal. What it adds is a functional *form*: two laws can share the same physical state as minimiser while differing completely in their stability, response, and behaviour under coarse-graining. A statistical law at representation r is a functional $J_r[\rho]$ whose minimisation selects the physical state; the object that can survive vertical changes of description is the functional form, not merely the selected state.

The question is:

> *What scalar comparison between finite descriptions survives staged loss of information?*

Four requirements on a scalar $\mathcal{I}(p\|q)$ express what any reasonable comparison between finite representations must respect.

Relabelling invariance. The comparison cannot depend on arbitrary names of states. Bijections of the state space leave $\mathcal{I}$ unchanged.

Atomwise locality. Disjoint alternatives contribute separately: $\mathcal{I}(p\|q) = \sum_i \ell(p_i, q_i)$ for a common local density ℓ.

Neutral refinement. Splitting an alternative into subalternatives with the same likelihood ratio p_i/q_i and no new visible distinction leaves $\mathcal{I}$ unchanged.

Staged elimination. Eliminating hidden distinctions in one step or in several must give the same total accounting:

$$\begin{aligned}\mathcal{I}_X(p_X\|q_X) &= \mathcal{I}_Y(p_Y\|q_Y)\\ &+\mathbb{E}_{p_Y}\mathcal{I}_{X_y}\left(p_{X|Y}(\cdot|y)\|q_{X|Y}(\cdot|y)\right).\end{aligned} \tag{4}$$

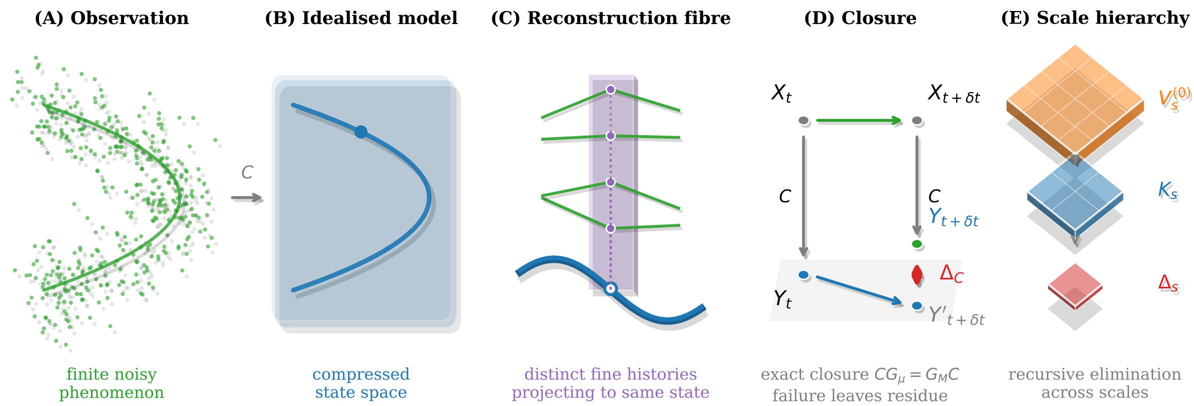

FIG. 2. **Finite law-form under change of description.** *(A)* Finite, noisy phenomenon: data lying near an idealised curve in the fine state space. *(B)* Coarse-graining C compresses it to a smaller state space X_M; the compressed curve is the idealised model. *(C)* Reconstruction fibre: each coarse state admits many fine histories; the fibre $C^{-1}(m)$ collects all of them. *(D)* Closure residue: evolving at the fine scale and coarse-graining yields $Y_{t+\delta t}$ (green); coarse-graining first and evolving coarsely yields $Y'_{t+\delta t}$ (grey); the gap $\Delta_C = CG_\mu - G_M C$ (red) is the determinate closure failure. *(E)* Scale triage: lower-scale structure distributes into null potential $V_s^{(0)}$ (amber), visible complexity K_s (blue), and closure residue Δ_s (red). See Sections III–IV.

The staged-elimination requirement is substantive. Any scalar failing (4) assigns different statistical content depending on the order in which hidden structure is discarded—which is incompatible with that structure being genuinely irrelevant.

Result 1. *For finite positive-support distributions, relabelling invariance, atomwise locality, neutral refinement, and staged elimination together force*

$$\mathcal{I}(p\|q) = C\sum_i p_i \log\frac{p_i}{q_i}. \tag{5}$$

Proof in Appendix A. Relative entropy is not imported from information theory. It is the unique scalar bookkeeping compatible with staged finite description. Every other continuous comparison violates the chain rule (4) by an $O(1)$ amount on the simplest finite examples (Fig. 3).

III. FREE ENERGY, NULL LAWS, AND INHERITED POTENTIALS

Real descriptions retain not just distributions but observable constraints. Visible constraints are the quantities the finite description chooses to keep. If the retained observables are A_a, then enforcing or biasing their values introduces conjugate coefficients g_a, giving

$$K_s = \sum_a g_a A_a. \tag{6}$$

The observables A_a name distinctions the representation can make; the coefficients g_a are the costs of making those distinctions matter. Thus K_s is the cost function of retained complexity. The null law q_s is the structure inherited for free at scale s, while K_s prices the additional complexity retained above that background. This is the usual Legendre structure of statistical mechanics: energy is paired with inverse temperature, particle number with chemical potential, magnetisation with magnetic field. The same logic applies to any retained coarse variable.

Adding visible constraints $\langle A_a\rangle_p = a_a$ to the KL law-content of Result 1 yields the forced law-functional.

Result 2. *For finite states with compatible support and $M > 0$, staged-elimination law-content combined with visible constraints gives the variational family*

$$\mathcal{F}_s[p] = \langle K_s\rangle_p + M D_{\mathrm{KL}}(p\|q_s), \tag{7}$$

where $K_s = \sum_a g_a A_a$ is the cost function of retained visible complexity.

The free-energy functional trades two costs: (i) the expected cost of retained visible complexity; (ii) the cost of departing from the recursively generated null law.

Proof and contraction-closure property in Appendix A. The free-energy form is not borrowed from thermodynamics; thermodynamics is recognised as an instance of this forced form once a heat bath is identified as the hidden refinement.

The distribution q_s appearing in (7) is not a reference measure chosen externally. At each scale s, there is a determinate answer to the question:

> *What is the minimally committed law compatible with the current partition, before any additional resolved structure is imposed?*

The answer is the *null law*: the law generated recursively by the coarse-graining itself. Let C be the coarse-graining from scale $s - ds$ to s, and let q_{s-ds} be the null

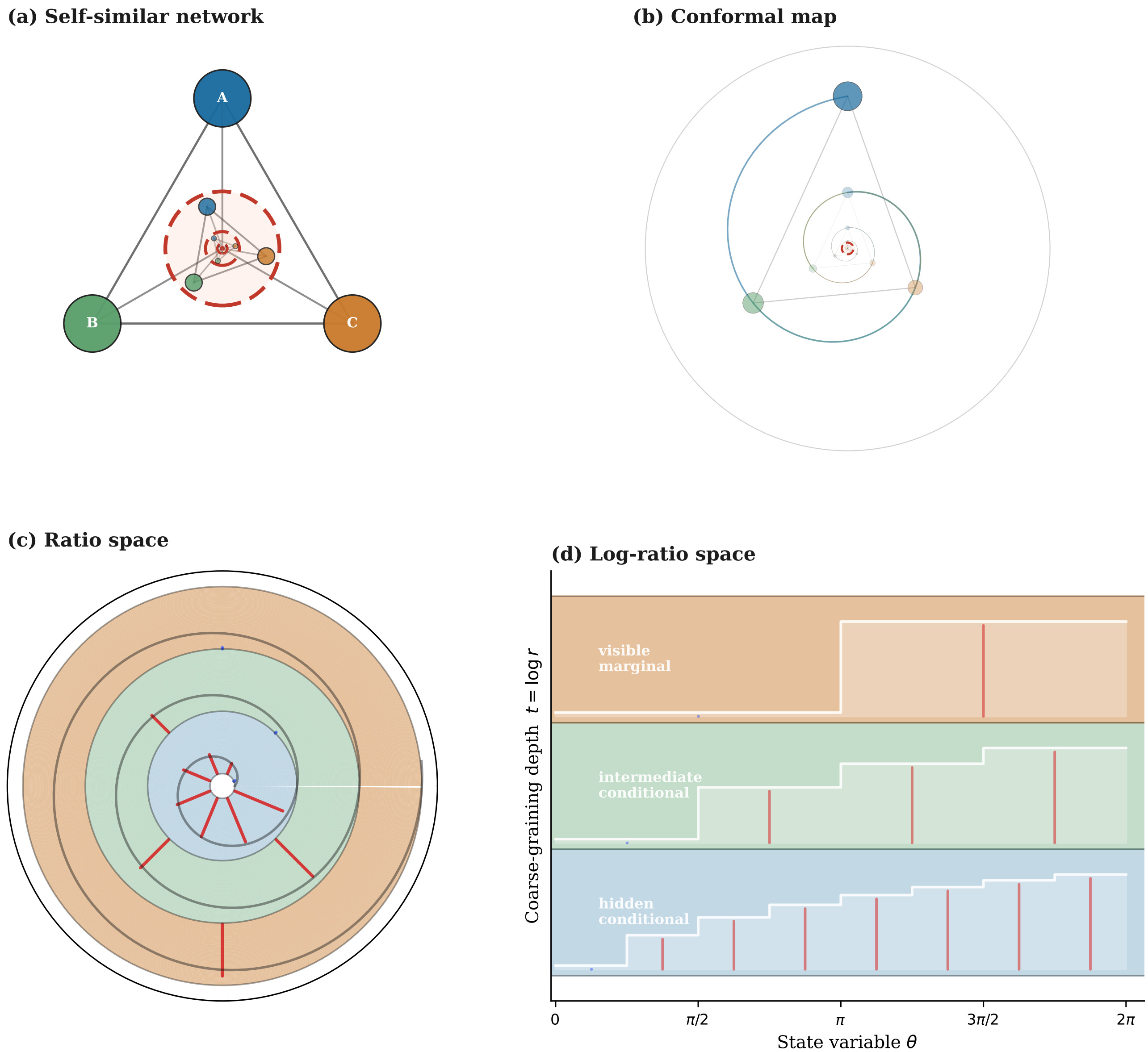

FIG. 3. **The logarithm is forced by staged elimination.** *(a)* Self-similar network: the wheel graph embeds its own hub at smaller scale, so ratio self-similarity is built into the structure. *(b)* Conformal log-spiral: the map $w = \log z$ converts multiplicative nesting into additive periodicity; each revolution of the spiral carries the same ratio structure. *(c)* Ratio space: probability ratios p_i/q_i for the rank ensemble $P_I(n) \propto n^{-I}$ displayed in polar coordinates, one ring per coarse-graining level. Red bars indicate $p_i/q_i > 1$; blue bars $p_i/q_i < 1$. *(d)* Log-ratio space: $\log(p_i/q_i)$ as a function of state variable θ, one band per level (fine, intermediate, visible). Logarithms add across levels: $\log r = \log r_Y + \log r_{X|Y}$, the chain rule (4). The additivity condition $\phi(ab) = \phi(a) + \phi(b)$ forces $\phi(a) = C \log a$, selecting the KL divergence uniquely. Derivation in Appendix A.

law at the finer scale. Pushing it forward and projecting onto the admissible class $\mathcal{N}_s$ gives

$$q_s = \arg\min_{q \in \mathcal{N}_s} D_{\mathrm{KL}}(C_* q_{s-ds} \| q)\,. \tag{8}$$

The null law is not a prior. It is the law obtained when the only retained constraints are those required to define the present representation; everything else has been discarded.

Define the *inherited null potential*

$$V_s^{(0)} = -M \log q_s \quad \text{(up to additive constant)}, \tag{9}$$

so that the total effective law at scale s is

$$\gamma_s \propto \exp\left[-\frac{K_s + V_s^{(0)}}{M}\right]. \tag{10}$$

The null potential is what the lower-scale law becomes when it has been absorbed into the self-consistent background of the current description. The kinetic-potential split—transition geometry versus occupancy weighting relative to the null law—is induced by this self-consistency, not assumed as a primitive mechanical dichotomy. The derivation is in Appendix B.

As scale advances from $s-ds$ to s, lower-scale structure triages into three branches:

$$\text{lower-scale law} \longrightarrow \begin{cases} V_s^{(0)} = -M \log q_s, & \text{null potential,} \\ K_s, & \text{visible complexity,} \\ \Delta_s, & \text{closure residue.} \end{cases} \tag{11}$$

This bookkeeping identity is the spine of the paper.

Compression constraints are a special case of this logic. A localised excitation is a compressed pattern of lower-scale structure that remains identifiable after coarse-graining. If the finite description retains a variable measuring the persistence of that compression, the conjugate coefficient entering K_s is the cost of maintaining it. When such a cost closes as a scalar, stable, transportable feature of the effective description, it has the structural role of a mass-like parameter. In this limited sense, mass is compression after closure: not compression itself, but the cost of maintaining a self-closing compression against the null law. The familiar local form of mass terms belongs to the path/action version of the theory, where the differential cost of retained complexity must be derived and then expanded in the appropriate small-deformation limit.

The next section develops the third branch.

IV. CLOSURE RESIDUE AND EFFECTIVE PHYSICS

Section III identified three branches of the triage (11); this section develops the third.

When a fine law is pushed forward through a coarse-graining, the result may not be representable exactly in the coarse law-form. The mismatch is the closure residue

$$\Delta_C = CG_\mu - G_M C, \tag{12}$$

already introduced in Section I. The question is whether this mismatch accumulates arbitrarily or composes with structure.

Result 3. *For a scale tower*

$$X_0 \xrightarrow{C_{01}} X_1 \xrightarrow{C_{12}} X_2, \tag{13}$$

with one-step residues $\Delta_{C01} = C_{01}G_0 - G_1C_{01}$ *and* $\Delta_{C12} = C_{12}G_1 - G_2C_{12}$, *the two-step closure residue satisfies*

$$\Delta_{C02} = C_{12}\,\Delta_{C01} + \Delta_{C12}\,C_{01}. \tag{14}$$

Proof in Appendix C. The cross term $C_{12}\,\Delta_{C01}$ is the key feature: closure residue is *transported* by subsequent coarse-grainings, not accumulated as additive noise. Residue at one scale enters the two-step mismatch only after being acted on by the next coarse-graining map. This coherent composition is what allows residue to become law-like effective structure rather than arbitrary fluctuation.

The triage (11) now acquires a classification. Seven modes of unabsorbed closure residue arise, ordered by algebraic character. The first—absorbable scalar residue—contracts under scale integration to a correction in the effective coupling constant: this is renormalisation-group flow, the most familiar mode. Transition residue is kinetic, appearing as friction or corrections to transition rates in open systems. Nonlocal residue persists across time and generates the Nakajima–Zwanzig memory kernel: the coarse system cannot be Markovian because its past is encoded in the hidden variables. Incoherent unresolved residue is noise and fluctuating forces. Coherent unresolved residue survives coarse-graining with internal phase relations intact, appearing as an emergent order parameter or new field. Reconstruction residue is gauge freedom: the ambiguity in how fine states are lifted back through the coarse-graining fibre. Finally, spectral residue arises when incompatible coarse-grainings produce noncommuting projectors, even from purely classical averaging. Effective physics is structured closure residue. Full classification and the two-mode open-system example are in Appendices C and D.

V. EQUILIBRIUM, EMERGENCE, AND CONSERVED STRUCTURE

The first two branches of the triage (11)—null potential and visible complexity—acquire concrete physical content here. Equilibrium is the state in which both branches close without residue; symmetry and conservation determine the algebra of what the null-potential branch inherits across scale transitions.

Equilibrium

A state is stationary when $G_s\gamma_s = 0$. But stationarity is not equilibrium. A driven steady state is stationary within a representation while maintaining itself through hidden fluxes that would be visible at a finer description.

Equilibrium requires in addition that the stationary law closes:

$$G_s\gamma_s = 0 \qquad \text{and} \qquad \Delta_C\,\gamma_s = 0. \tag{15}$$

Equilibrium is the closed sector of finite description: the part that reproduces itself without leaving closure residue under the relevant scale change.

> *Nonequilibrium is not merely motion. It is persistent closure residue.*

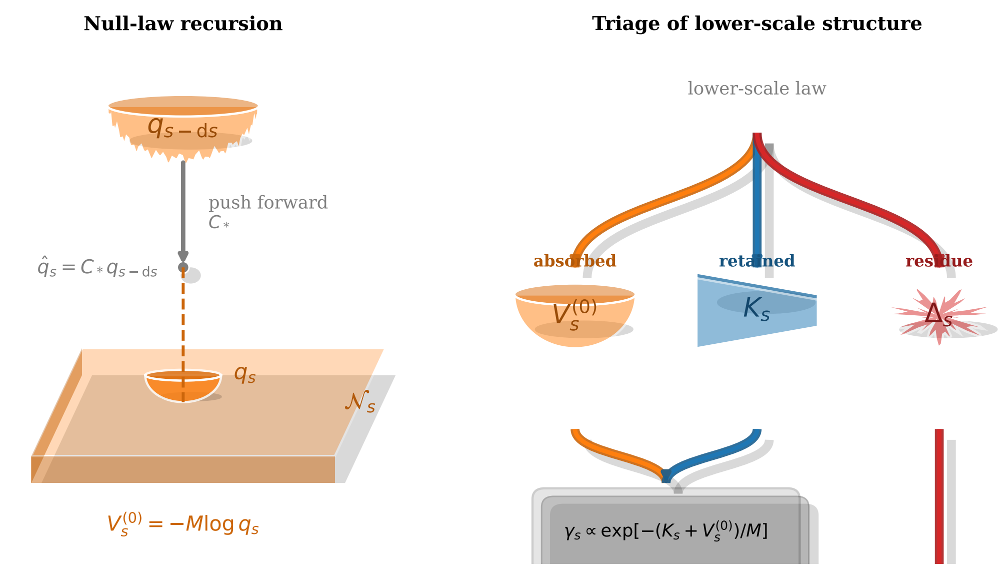

FIG. 4. **Null law recursion and residue triage.** *Left:* The finer null law q_{s-ds} is pushed forward through C to give $\hat{q}_s = C_* q_{s-ds}$, then projected onto the admissible class $\mathcal{N}_s$ to give q_s (amber, Eq. 8). The inherited null potential $V_s^{(0)} = -M \log q_s$ (amber) is the lower-scale law structure absorbed into the current-scale background. The visible deformation K_s (blue) is resolved structure remaining after absorption; their sum $K_s + V_s^{(0)}$ is the total effective generator. *Right:* The triage of lower-scale law (Eq. 11). Every part of the lower-scale law is either absorbed (amber), retained (blue), or left as closure residue (red). Recursive null-law derivation in Appendix B.

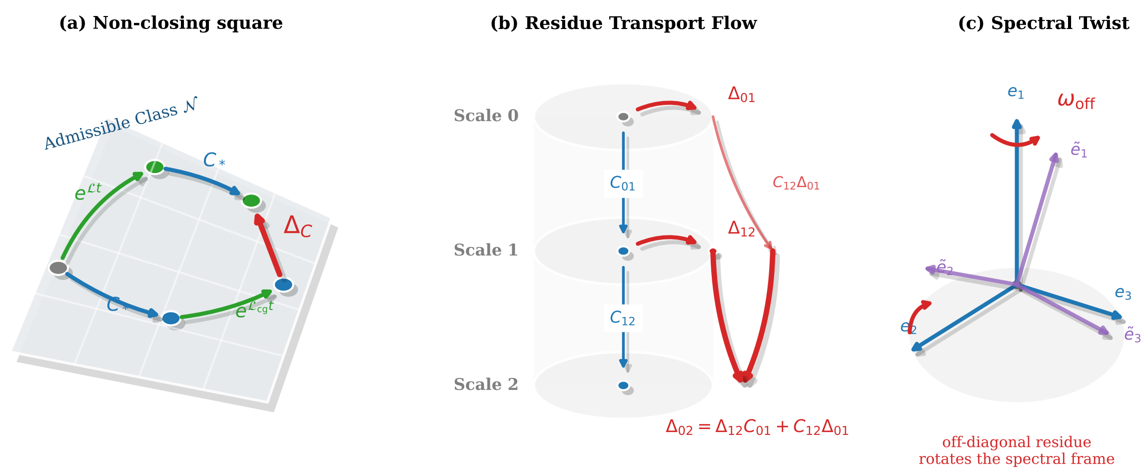

FIG. 5. **Closure residue: non-closing square, composition, and spectral limit.** *Left:* Two paths from (X_μ, t) to the coarse level at $t + \delta t$: evolve finely then coarsen (green/blue), versus coarsen first then evolve coarsely (orange). The gap is $\Delta_C = CG_\mu - G_M C$ (red). Exact closure is the commuting square; closure residue is the failure. *Centre:* Three-scale composition identity $\Delta_{C02} = C_{12}\Delta_{C01} + \Delta_{C12}C_{01}$ (Result 3). The cross term shows residue is transported by the next coarse-graining, not accumulated as additive error. Numerical verification: residual 3.93×10^{-17}. *Right:* In the infinitesimal limit the finite residue Δ_C yields the residue density ω_s (Eq. 22, Section VI). Off-diagonal ω_s drives spectral-frame rotation; diagonal ω_s renormalises generator/potential data. Full derivation in Appendix F.

Entropy measures the hidden structure that can be *absorbed* into the null law q_s without leaving closure residue. When the entropy of the hidden conditional is the only contribution to the change in null potential, the system closes exactly. Unabsorbable structure—memory, correlated residues, coherent modes—leaves $\Delta_C \neq 0$. Details in Appendix E.

Symmetry and emergence

To coarse-grain is to choose a symmetry structure: to decide which distinctions are gauge-like from the coarser point of view and which remain physical.

A fine symmetry g_μ transports to a coarse symmetry

g_M only if

$$Cg_\mu = g_M C. \tag{16}$$

Three cases arise: the symmetry transports and remains a symmetry; the transformation satisfies $Cg_\mu = C$ and becomes a gauge redundancy at the coarse level; or no g_M closes the diagram (failed closure). The third case is *emergence*: the residue of the failed symmetry closure appears as new structure at the coarser scale. Emergence is not vague novelty—it is the specific algebraic failure of the commuting square, measured by the closure residue of the symmetry.

Conserved quantities

Define the *dual closure residue*

$$\Delta_C{}^\dagger = G_\mu^\dagger C^* - C^* G_M^\dagger, \tag{17}$$

where $G^\dagger$ acts on observables rather than states. If $G_M^\dagger I_M = 0$ (that is, I_M is a macro integral of motion), then

$$G_\mu^\dagger C^* I_M = \Delta_C{}^\dagger I_M. \tag{18}$$

A macro conservation law is *inherited* when $\Delta_C{}^\dagger I_M = 0$: the lifted observable is also conserved at the fine level. It is *emergent or approximate* when $\Delta_C{}^\dagger I_M \neq 0$: the conservation holds at the coarse level through a cancellation that fails to lift. Dual-residue derivations and the connection to integrable systems are in Appendix E.

VI. INFINITESIMAL RESIDUE AND QUANTUM-LIKE CLOSURE

The spectral residue mode—the rightmost in the classification of Section IV—is the subject of this section. What does the closure residue look like in the infinitesimal scale limit, and what fixed-point behaviour can it exhibit?

Let the coarse-graining between adjacent scales be

$$C_{s\to s+\delta s} = I + \delta s\, L_s + O(\delta s^2), \tag{19}$$

and let the generator vary as

$$G_{s+\delta s} = G_s + \delta s\, \dot{G}_s + O(\delta s^2). \tag{20}$$

A direct expansion of the closure residue gives

$$\Delta_C{}^{s\to s+\delta s} = \delta s\big([L_s, G_s] - \dot{G}_s\big) + O(\delta s^2). \tag{21}$$

The raw residue vanishes with δs, but its ratio to δs remains finite. Define the *residue density*

$$\omega_s = [L_s, G_s] - \dot{G}_s. \tag{22}$$

The fixed-point object is ω_s, not the raw finite residue.

For a non-degenerate spectrum $G_s = \sum_i \lambda_i \Pi_i$, the off-diagonal residue drives spectral-frame rotation:

$$\Pi_j (D_s \Pi_i) \Pi_i = -\frac{\Pi_j\, \omega_s\, \Pi_i}{\lambda_i - \lambda_j}, \tag{23}$$

where $D_s = \partial_s - \mathrm{ad}(L_s)$ is the covariant scale-derivative. Diagonal residue $(\omega_s)_{ii}$ renormalises eigenvalues and can be absorbed into K_s or $V_s^{(0)}$. Off-diagonal residue $(\omega_s)_{ij}$, $i \neq j$, rotates spectral projectors and *cannot* be absorbed into any scalar potential.

For the two-level case $G_s = \frac{\Delta}{2}\sigma_z$, $\omega_s = \kappa\, \sigma_x$, one obtains

$$\lim_{\delta s\to 0} \frac{[\Pi_+(s),\, \Pi_+(s+\delta s)]}{\delta s} = -i\, \frac{\kappa}{\Delta}\, \sigma_y. \tag{24}$$

Derivation in Appendix F. The commutator density is finite even as the raw commutator vanishes.

Define the dimensionless spectral residue ratio

$$\chi_s = \frac{\text{off-diagonal residue density}}{\text{spectral gap}} = \frac{\kappa}{\Delta} \quad \text{(two-level case).} \tag{25}$$

Two regimes follow from the scale evolution of χ_s:

$$\chi_s \to 0: \quad \text{classical closure.} \tag{26}$$

Spectral frames become asymptotically compatible; a single joint distribution assigns probabilities to all contexts simultaneously.

$$\chi_s \to \chi^* \neq 0: \quad \text{self-similar spectral residue.} \tag{27}$$

No scale transition can be made invisible; the spectral-frame rotation rate is finite and scale-stable.

Ordinary quantum mechanics is not derived here. The claim is that quantum theory has the structural form expected when spectral closure residue becomes self-similar, central, and universal. In that regime, an operational residue scale $\hbar_{\mathrm{op}}(s, \rho, C, R)$ governs the stable spectral residue; the quantum-mechanical limit is

$$\hbar_{\mathrm{op}}(s, \rho, C, R) \to \hbar\, I. \tag{28}$$

Extended treatment of self-similar residue dynamics, generalised quantum-like regimes, and the nonclaims of this identification are in Appendix G.

VII. NEURAL SCALING LAWS AS CLOSURE-RESIDUE DECAY

The closure-residue branch of the triage has a direct empirical target: modern neural network training, where the statistical law, the representation scale, and the residue are all measurable.

Consider a data distribution p_{data} and a parametrised model distribution p_θ. The standard training loss is the cross-entropy,

$$\mathcal{L}(\theta) = -\mathbb{E}_{x\sim p_{\mathrm{data}}} \log p_\theta(x). \tag{29}$$

This decomposes into the irreducible entropy $H(p_{\mathrm{data}})$ of the data source and a KL divergence,

$$\mathcal{L}(\theta) - H(p_{\mathrm{data}}) = D_{\mathrm{KL}}(p_{\mathrm{data}} \| p_\theta). \tag{30}$$

The irreducible entropy is the null statistical content of the data source, fixed independently of the model. The excess loss is the scalar representational residue: the part of the data law not generated by the model family.

Scaling model size, data, and compute changes the admissible representation. Let $s = \log N$, where N is a representation-scale variable such as parameter count. If the representation flow is locally self-similar, residue mode k evolves as $a_k(s) \sim e^{-\lambda_k s}$, giving $a_k(N) \sim N^{-\lambda_k}$. The slowest-decaying mode dominates,

$$\mathcal{L}(N) - \mathcal{L}_\infty \sim AN^{-\alpha}. \tag{31}$$

Power-law scaling is the signature of self-similar residue decay, not an empirical coincidence.

There are at least two leading residues: $\Delta_{\mathrm{cap}}(N)$, the capacity residue controlled by model size; and $\Delta_{\mathrm{samp}}(D)$, the sampling residue controlled by dataset size. Therefore,

$$\mathcal{L}(N, D) - \mathcal{L}_\infty \approx AN^{-\alpha} + BD^{-\beta} \tag{32}$$

plus interaction terms. Under a compute constraint $C \sim ND$, compute-optimal scaling balances the marginal reduction of each residue: neither mode should dominate.

The smooth power law is the self-similar decay of residue modes with no residue-class transitions. Deviations from smoothness mark residue changing class: absorbable scalar residue yields smooth coupling flow; nonlocal residue corresponds to context dependence; coherent unresolved residue marks circuit or feature formation; incoherent residue appears as noise; capacity-pressured coherent residue manifests as superposition. Induction heads are a closure event: history-dependent residue becomes an internal transition operator. Superposition is coherent unresolved residue compressed into too few visible dimensions.

Defining a smooth scaling baseline and extracting the residual,

$$R(N, D) = \mathcal{L}_{\mathrm{obs}}(N, D) - \mathcal{L}_{\mathrm{fit}}(N, D), \tag{33}$$

the framework predicts that circuit formation, feature splitting, induction-head emergence, and superposition reorganisations should correlate with structured features of $R(N, D)$—extrema, knees, or exponent changes—rather than occurring randomly along the scaling curve. Neural networks are not special physical systems; they are unusually well-instrumented finite theories in which the dynamics of representation is directly measurable. (Appendix H gives the full treatment.)

VIII. DISCUSSION

Finite descriptions are the normal case in physics, not a limitation to be apologised for. Every physical variable is a coarse image of phenomena richer than itself; every equation governs one level in a hierarchy that extends both above and below. The question this paper has pursued—what consistency conditions does changing description impose on a law-form?—is therefore not a philosophical aside but a constraint operating in every equation in use.

The answer takes the form of a forced triage (11). Lower-scale structure distributes into three determinate branches when a scale transition is made: what closes exactly becomes the null law, generating the recursively self-consistent background at the coarser scale; what remains visible is priced by the cost-function K_s; and what cannot be absorbed or visibly retained is the closure residue, whose composition identity (14) guarantees coherent, transportable structure rather than arbitrary noise. The framework is in this sense one of legal relativity: every law has authority relative to a partition—its jurisdiction—and the triage is the transformation rule as jurisdictions change. The KL functional is the covariant invariant; generators and null laws are the jurisdiction-specific content.

The modes of that residue are the modes of effective physics. Renormalisation-group flow is absorbable scalar residue. Memory kernels are nonlocal residue. Emergent order parameters are coherent unresolved residue. Gauge freedom is reconstruction residue. Entropy is the scalar bookkeeping of structure that closes exactly, leaving no residue. None of these need to be assumed; they are the algebra of what closure failure can look like, forced by the triage condition and the coherent composition of residue across scales.

The most striking consequence emerges from the reflexive character of the theory itself. A theory of Ω is a finite register $R \subset \Omega$, subject to the same triage as everything else it describes. When the consistency conditions close the loop on the observer—when the act of distinguishing outcomes is itself a coarse-graining subject to the same calculus—self-consistent closure requires residue. The observer has a finite partition; its distinguishable elements are exactly that, and no more; and no finite description can fully absorb the consequences of applying its own coarse-graining to itself. What quantum mechanics reports is that this loop has a stable, rigid closure: the scalar, central spectral fixed point. Its existence is evidence that the loss is irreducible and structured, not arbitrary—the observer cannot distinguish everything precisely because its own partition is finite and reflexive, and the Hilbert-space formalism is the algebra that closure forces on any description that accounts for this fact.

Four sharp mathematical frontiers remain. First, the KL and free-energy uniqueness theorems are proved for finite positive-support descriptions; extending them to infinite or continuous descriptions requires additional regularity. Second, the null-law projection is well-defined for convex, closed admissible classes; the non-convex case—symmetry breaking, phase coexistence—requires a branch-selection principle. Third, the spectral fixed-point characterisation identifies the structural condition for quantum-like closure; the full analysis of the reflexive observer loop, including what determines the algebra of distinguishable events, is the next natural problem. Fourth, the neural scaling prediction correlates residue-class transitions with observable features of the loss curve; establishing causal links requires controlled perturbation experiments. These are not qualifications but the natural next problems the framework opens.

Appendix A: KL and free-energy derivations

This appendix proves Results 1 and 2 and establishes contraction closure of the free-energy family.

1. Local ratio form

Proposition 1 (Local ratio form)**.** *For finite positive distributions p, q satisfying relabelling invariance and atomwise locality, the scalar law-content has the form*

$$\mathcal{I}(p\|q) = \sum_i p_i\, \varphi\left(\frac{p_i}{q_i}\right) \tag{A1}$$

for a continuous function φ on $\mathbb{R}_+$.

Proof. Atomwise locality gives $\mathcal{I}(p\|q) = \sum_i \ell(p_i, q_i)$ with a common density ℓ. For fixed ratio $r = a/b$ define $g_r(t) = \ell(rt, t)$. Neutral refinement: splitting a cell of q-mass $t_1 + t_2$ into two neutral subcells with q-masses t_1, t_2 and the same ratio r gives

$$g_r(t_1 + t_2) = g_r(t_1) + g_r(t_2).$$

Continuity forces $g_r(t) = t\, g_r(1)$, so $\ell(a, b) = a\, \varphi(a/b)$ with $\varphi(r) = g_r(1)/r$. □

2. KL uniqueness under staged elimination

Proposition 2 (KL uniqueness)**.** *For finite positive distributions satisfying the local-ratio form of Proposition 1, the staged-elimination chain rule* (4) *forces*

$$\mathcal{I}(p\|q) = C \sum_i p_i \log \frac{p_i}{q_i}. \tag{A2}$$

Proof. Refine coarse state i into conditional alternatives j: set $p_{ij} = p_i r_{j|i}$, $q_{ij} = q_i s_{j|i}$, $a_i = p_i/q_i$, $b_{j|i} = r_{j|i}/s_{j|i}$. The direct fine content is $\sum_{ij} p_i r_{j|i}\, \varphi(a_i b_{j|i})$. The chain rule requires this to equal

$$\sum_i p_i\, \varphi(a_i) + \sum_i p_i \sum_j r_{j|i}\, \varphi(b_{j|i})$$

for all positive refinements.

Step 1: $\varphi(1) = 0$. Apply neutral refinement: set $b_{j|i} = 1$ for all j, i (equal likelihood ratios in every conditional cell). Then $a_i b_{j|i} = a_i$, so both sides reduce to $\sum_i p_i \varphi(a_i)$, which forces $\sum_i p_i \sum_j r_{j|i}\, \varphi(1) = 0$ for every admissible p, r. Choosing any nondegenerate p, r gives $\varphi(1) = 0$.

Step 2: Isolate arbitrary a, b. Fix two arbitrary positive reals a, b. Choose a coarse description with a single nontrivial state $i = 1$ with ratio $a_1 = a$. Refine state 1 into two sub-states $j = 1, 2$ with conditional weights $r_{1|1} = \epsilon$, $r_{2|1} = 1 - \epsilon$ for small $\epsilon > 0$, and conditional ratios $b_{1|1} = b$, $b_{2|1} = 1$. All other coarse states are refined neutrally.

Step 3: Apply chain rule and solve. The chain rule equation for this refinement reads, after cancelling neutral terms,

$$\epsilon\, \varphi(ab) + (1 - \epsilon)\, \varphi(a) = \varphi(a) + \epsilon\, \varphi(b) + (1 - \epsilon)\, \varphi(1).$$

Using $\varphi(1) = 0$ and cancelling $(1 - \epsilon)\varphi(a)$ from both sides:

$$\epsilon\, \varphi(ab) = \epsilon\, \varphi(a) + \epsilon\, \varphi(b).$$

Dividing by $\epsilon > 0$ gives

$$\varphi(ab) = \varphi(a) + \varphi(b) \quad \text{for all } a, b \in \mathbb{R}_+.$$

Since a and b were arbitrary, this holds on all of $\mathbb{R}_+$. Continuity forces $\varphi(a) = C \log a$. Positivity forces $C \geq 0$; nontriviality fixes $C > 0$. □

The results align with the Csiszár characterisation of f-divergences [1] and the functorial characterisations of [2, 3]. The setting here is strictly finite and positive-support; extensions to measurable spaces require domain and regularity conditions beyond the scope of this paper.

3. Free-energy law-form

Proposition 3 (Free-energy law-form)**.** *For finite states with compatible support and $M > 0$, the unique staged-elimination law-content combined with visible constraints $\langle A_i \rangle_p = a_i$ gives the variational family*

$$\mathcal{F}_s[p] = \langle K \rangle_p + M D_{\mathrm{KL}}(p\|q) \tag{A3}$$

with generator $K = \sum_i g_i A_i$ and minimiser $p^*(x) \propto q(x) \exp[-K(x)/M]$.

Proof. With KL law-content and Lagrange multipliers g_i and α, the Euler equation gives $M(\log p - \log q) + K + \alpha = 0$, hence $p^* \propto q \exp[-K/M]$. Substituting back gives minimum free energy $\mathcal{F}^* = -M \log Z$ with $Z = \sum_x q(x) e^{-K(x)/M}$. □

4. Contraction closure

Proposition 4 (Contraction closure)**.** *For a finite coarse map $\pi : X \to Y$ with decomposition $q_X(x) = q_Y(y)\, q_{X|Y}(x|y)$, the fibre contraction of the free-energy functional is again a free-energy functional:*

$$\inf_{\pi_* p_X = p_Y} \mathcal{F}_X[p_X] = \sum_y p_Y(y)\, E_{\mathrm{eff}}(y) + M D_{\mathrm{KL}}(p_Y\|q_Y), \tag{A4}$$

where

$$E_{\mathrm{eff}}(y) = -M \log \sum_{x \in X_y} q_{X|Y}(x|y)\, e^{-E_X(x)/M}. \tag{A5}$$

Proof. Apply the KL chain rule to decouple the conditional minimisation over fibres. At fixed p_Y, minimising over $p(\cdot|y)$ for each y gives the Gibbs conditional

$$p^*(x|y) = \frac{q_{X|Y}(x|y)\, e^{-E_X(x)/M}}{\sum_{x' \in X_y} q_{X|Y}(x'|y)\, e^{-E_X(x')/M}}.$$

Substituting gives the effective energy $E_{\text{eff}}(y)$ as stated, and the remaining p_Y-minimisation is exactly a free-energy problem with generator E_{eff}. □

Contraction closure is the precise content of vertical covariance of law-form: eliminating hidden refinements does not change the *kind* of law, only the generator and its coordinates. Ordinary renormalisation-group coupling flow is the induced flow of K within this invariant free-energy family (Appendix D).

Appendix B: Null laws and kinetic/potential split

1. Recursive null-law projection

The null law at each scale is defined recursively by the coarse-graining operation itself. Let $\mathcal{N}_s$ denote the admissible class at scale s: the set of distributions compatible with the partition and its defining constraints, and no others.

Definition 1 (Null law). *The null law q_s at scale s is*

$$q_s = \arg\min_{q \in \mathcal{N}_s} D_{\text{KL}}(C_* q_{s-ds} \| q)\,, \tag{B1}$$

where $C_* q_{s-ds}$ is the push-forward of the finer null law through the coarse-graining C.

At a self-similar fixed point $q^* = \mathcal{P}(C_* q^*)$, where $\mathcal{P}$ denotes projection onto $\mathcal{N}_s$, the null law is self-consistently the baseline produced by the scale's own coarse-graining. The null law at the finest scale (the ground-level description) is the uniform distribution over the primitive alternatives, which is the only distribution in $\mathcal{N}$ when no constraints are available.

The null law is not a prior in the Bayesian sense. It is not chosen before evidence or placed externally by an agent. It is the output of the same projection operation that defines the scale.

2. Inherited null potential

Write the null law in exponential form:

$$q_s(x) = \frac{1}{Z_s^{(0)}} \exp\Big[-V_s^{(0)}(x)/M\Big]\,. \tag{B2}$$

The *inherited null potential* $V_s^{(0)} = -M \log q_s$ encodes the accumulated effect of all lower-scale laws that have been successfully absorbed into the current-scale background through repeated projection.

At each new scale:

1. The finer null law q_{s-ds} is pushed forward to $\hat{q}_s = C_* q_{s-ds}$.
2. $\hat{q}_s$ is projected onto $\mathcal{N}_s$ to give q_s.
3. $V_s^{(0)} = -M \log q_s$ is the null potential at the new scale.

Any lower-scale structure that cannot be represented within $\mathcal{N}_s$ after this projection contributes to the closure residue Δ_s, not to the null potential.

The compact identity is:

$$\Delta V_s^{(0)} = -M \log Z_{\text{conditional}} = \langle K_{\text{hidden}} \rangle + M H_{\text{conditional}}, \tag{B3}$$

where $H_{\text{conditional}}$ is the Shannon entropy of the hidden conditional. Entropy is the scalar bookkeeping of structure that *can* be absorbed into the null potential without closure residue.

3. Path formulation and the kinetic/potential split

The kinetic-potential split follows from applying the free-energy construction to path-space rather than instantaneous state-space.

For a path $x(t)$ over the interval $[0, T]$, the effective action selecting physical trajectories decomposes as

$$\mathcal{A}_s[\text{path}] = T_s[\text{path}] + V_s[\text{path}], \tag{B4}$$

where:

- $T_s[\text{path}]$ is the transport cost of the trajectory given the admissible null dynamics: the kinetic contribution from allowed transitions of the represented variables.
- $V_s[\text{path}]$ is the occupancy cost relative to the null law: the integrated KL weight of the trajectory against the scale-s background.

Both contributions arise from the same free-energy construction applied to measures over paths. The split is not assumed as a primitive mechanical dichotomy ("kinetic" and "potential" energy as independent inputs). It is induced by self-consistent coarse-graining.

Kinetic component. The null dynamics determines which transitions between represented states are admissible and at what rate. The transport cost T_s measures deviation from those admissible null transitions. In a Hamiltonian system this recovers the familiar kinetic energy (cost of motion given the metric); in an open or stochastic system it gives the appropriate path-space transport term.

Potential component. The total effective potential is

$$V_s^{\text{eff}}(x) = V_s^{(0)}(x) + K_s(x), \tag{B5}$$

with inherited null potential $V_s^{(0)}$ and visible deformation K_s. Effective potentials produced by RG integration of hidden degrees of freedom are inherited null-potential components, not primitive energies.

Temperature. The scale M relates the two components: it is the ratio of potential to kinetic cost in the free-energy functional. In the standard thermodynamic context, $M = k_B T$. In more general settings, M is the scale parameter of the law-content.

Appendix C: Closure residue algebra

1. Composition proof

Proposition 5 (Closure-residue composition)**.** *For the scale tower* $X_0 \xrightarrow{C_{01}} X_1 \xrightarrow{C_{12}} X_2$ *with generators* G_0, G_1, G_2*, the two-step closure residue satisfies*

$$\Delta_{C02} = C_{12}\,\Delta_{C01} + \Delta_{C12}\,C_{01}. \tag{C1}$$

Proof. Expand $\Delta_{C02} = C_{12}C_{01}G_0 - G_2C_{12}C_{01}$ and add and subtract $C_{12}G_1C_{01}$:

$$\begin{aligned}\Delta_{C02} &= C_{12}(C_{01}G_0 - G_1C_{01}) + (C_{12}G_1 - G_2C_{12})C_{01} \\ &= C_{12}\,\Delta_{C01} + \Delta_{C12}\,C_{01}.\end{aligned}$$

□

The identity holds for linear generators on arbitrary vector spaces. For stochastic kernels it holds because they act linearly on signed measures. For differentiable nonlinear coarse maps and vector-field generators, the corresponding tangent-map identity is $\Delta_{C02}(x) = TC_{12} \cdot \Delta_{C01}(x) + \Delta_{C12}(C_{01}x)$. Numerical verification on 3×3 random matrices gives residual 3.93×10^{-17}.

Cocycle interpretation. The composition law (C1) has the structure of a 1-cocycle in the cochain complex of coarse-graining maps. Define the coboundary operator $(\delta R)_{02} = C_{12}R_{01} - R_{02} + R_{12}C_{01}$. The composition identity says that Δ_C is a closed 1-cochain: the residue distributes coherently over scale composition. This is why residue does not merely add as scalar noise: it is carried by the subsequent coarse-graining maps.

2. Full residue classification

Table I classifies the modes of closure residue $\Delta_C = CG_\mu - G_M C$ and the familiar structures they produce.

The triage (11) in the main text partitions lower-scale structure into absorbed (null potential), retained (visible deformation), and residue. The table above further partitions the residue branch according to the structure of Δ_C. Each row is not a hypothesis about what residue looks like; it is the characterisation of a specific algebraic property of the mismatch operator.

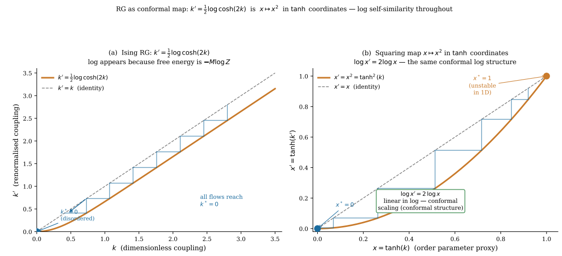

FIG. 6. **Ising RG as conformal map (appendix figure).** *Left:* Cobweb diagram for the 1D Ising recursion $k' = \frac{1}{2}\log\cosh(2k)$. *Right:* In tanh coordinates the recursion is the squaring map $x \mapsto x^2$, which is linear in log-coordinates. The logarithm appears because it is the generator flow inside the invariant free-energy form.

Appendix D: Worked examples

1. Ising decimation: RG as generator flow

Decimate the middle spin in a three-spin Ising block with generator $K(s_1, e, s_2) = -J(s_1e + es_2)$, retaining variables (s_1, s_2). The effective generator is given by contraction closure (Proposition 4):

$$K_{\text{eff}}(s_1, s_2) = -M\log\left[2\cosh\left(\frac{J(s_1+s_2)}{M}\right)\right]. \tag{D1}$$

Projecting onto the retained basis $\{1, s_1s_2\}$ gives the one-step coupling flow

$$J' = \frac{M}{2}\log\cosh\left(\frac{2J}{M}\right). \tag{D2}$$

For the full one-dimensional chain, repeated decimation gives

$$\tanh k' = \tanh^2 k, \qquad k = J/M, \tag{D3}$$

with all finite-temperature trajectories flowing to $k^* = 0$.

The free-energy law-form is preserved at every step. Only the coupling coordinate k changes. This confirms the contraction-closure property: RG is generator flow within the invariant free-energy family, not a modification of the family itself. The logarithm appears in (D1) for the same reason it appears in the law-content: it is the generator flow inside the invariant form.

In tanh coordinates $x = \tanh k$, the recursion reduces to the squaring map $x \mapsto x^2$: a linear (self-similar) map in log-coordinates, the same conformal scaling structure as the spiral in Fig. 3.

TABLE I. Classification of closure-residue modes and the familiar structures they produce. Each row corresponds to a distinct treatment of the mismatch $\Delta_C = CG_\mu - G_M C$.

Mode of Δ_C	Condition	Familiar structure
Zero	$\Delta_C = 0$	Autonomous classical closure
Negligible	$\|\Delta_C\|$ small	Effective classical theory
Absorbable scalar	reexpansion in retained basis	RG coupling flow
Transition	alters kinetic structure	Open-system friction
Nonlocal	no Markovian closure	Memory kernels (Nakajima-Zwanzig)
Incoherent	random at coarse scale	Noise, fluctuating forces
Coherent	structured at coarse scale	Emergent order parameter / field
Reconstruction	non-flat fibre transport	Gauge freedom, holonomy
Spectral	noncommuting fine/coarse frames	Spectral noncommutativity
History-dependent	non-Markovian	Influence functionals

2. Two-mode open system: nonlocal closure residue

Let the fine state be (u, v) with u retained and v hidden. Take the generator

$$G_\mu = \begin{pmatrix} 0 & a \\ -b & -\gamma \end{pmatrix}, \qquad \gamma > 0. \tag{D4}$$

The coarse map is $C = (1, 0)$. For any scalar macro-generator G_M,

$$\Delta_C = CG_\mu - G_M C = (-G_M,\ a). \tag{D5}$$

No choice of scalar G_M eliminates the a component when $a \neq 0$: this is nonlocal closure residue, mode 3 of Table I.

Solving for $v(t)$ and substituting into the u-equation gives

$$\dot{u}(t) = ae^{-\gamma t} v(0) - ab \int_0^t e^{-\gamma(t-s)} u(s)\, ds. \tag{D6}$$

The first term is transient noise from the hidden initial condition; the second is an exponential memory kernel. This is the Nakajima-Zwanzig structure [4, 5] at the level of the two-mode example.

In the scale-separated limit $\gamma \gg |\dot{u}/u|$, the kernel contracts:

$$\int_0^t e^{-\gamma(t-s)} u(s)\, ds \approx \frac{1}{\gamma}\, u(t), \tag{D7}$$

giving $\dot{u}(t) \approx -(ab/\gamma)\, u(t)$. The Markovian generator $G_M \approx -ab/\gamma$ is the scale-separated compression of the nonlocal residue into an absorbable scalar term. This illustrates mode 3 (nonlocal residue) transitioning to mode 1 (absorbable scalar) under scale separation: the microscopic mechanism behind nearly all effective dissipative dynamics.

The full Nakajima-Zwanzig identity is:

$$\frac{d}{dt} Px(t) = PLPx(t) + \int_0^t PLe^{(t-s)QL} QL\, Px(s)\, ds + PLe^{tQL} Qx(0), \tag{D8}$$

where P projects onto retained observables and $Q = 1 - P$. The first term is the closed retained generator; the integral is the memory kernel; the final term is noise from initial hidden-variable values. These correspond precisely to modes 1, 3, and 4 of the residue classification.

Appendix E: Equilibrium, symmetries, and conserved algebras

1. Stationarity versus equilibrium

The standard definition of equilibrium is stationarity: $G_s \gamma_s = 0$. This is necessary but not sufficient. A state can be stationary within a model while maintaining itself through hidden fluxes, unresolved memory, or scale-specific cancellations that do not survive a change of description.

In the present framework, equilibrium requires additionally that the stationary law closes without residue:

$$G_s \gamma_s = 0 \quad \text{and} \quad \Delta_C\, \gamma_s = 0. \tag{E1}$$

A driven steady state is stationary ($G_s \gamma_s = 0$) but not in equilibrium because it requires a continuous flow of hidden variables (heat baths, driving forces, environmental gradients) that would be visible at a finer description; this flow appears as $\Delta_C\, \gamma_s \neq 0$ at that finer level.

The null law q_s is the equilibrium law of the scale-defining constraints: it is the minimally informative stationary law compatible with the current partition. The Gibbs law $\gamma_s \propto \exp[-(K_s + V_s^{(0)})/M]$ is the equilibrium of the full set of retained constraints. Both are equilibria; the difference is which constraints are enforced.

2. Entropy as absorbable structure

When hidden structure is eliminated, it either becomes part of the null potential $V_s^{(0)}$ (absorbed into the background, contributing to thermodynamic work terms) or leaves closure residue Δ_s.

The absorbed part is the entropy of the hidden conditional:

$$\Delta V_s^{(0)} = -M \log Z_{\text{conditional}} = \langle K_{\text{hidden}} \rangle + M H_{\text{conditional}}, \tag{E2}$$

where $H_{\text{conditional}}$ is the Shannon entropy of the hidden variables conditional on the retained ones.

Entropy is therefore not a measure of ignorance in an epistemological sense. It is the scalar bookkeeping of structure that can be absorbed into the null potential without closure residue. Entropy derivations from maximum-entropy and large-deviation principles [6, 7] are special cases of this absorption construction.

3. Partition-relative symmetry

At a fixed partition, the theory has a family of horizontal symmetries: coordinate changes, basis rotations, canonical transformations. These preserve the set of distinctions the theory makes.

Changing the partition changes which symmetries exist. A coarse-graining C declares fine distinctions inside fibres to be invisible. Transformations g satisfying $Cg = C$ move within fibres and are gauge-like from the coarse point of view.

Definition 2 (Symmetry transport)**.** *A fine symmetry g_μ is partition-covariant if there exists g_M with $Cg_\mu = g_M C$. It is gauge-like from the coarse point of view if $Cg_\mu = C$. Closure fails if no g_M closes the commuting square; the symmetry is fibre-invisible or produces emergence at the coarser level.*

The three cases produce:

1. $Cg_\mu = g_M C$ for some g_M: the symmetry is transported symmetry and remains a symmetry of the coarser theory.

2. $Cg_\mu = C$: the transformation is fibre-invisible, a gauge redundancy at the coarse level.

3. No g_M makes the diagram commute: failed closure. The closure residue of the symmetry is emergent structure at the coarser scale.

Emergence is the specific algebraic failure of the commuting square, not a vague appeal to novelty.

Symmetry breaking versus symmetry emergence. Symmetry *breaking* refers to a symmetric law whose selected state is not symmetric. Symmetry *emergence* (case 3 above) refers to a fine symmetry that fails to transport to the coarser level. Both are possible simultaneously.

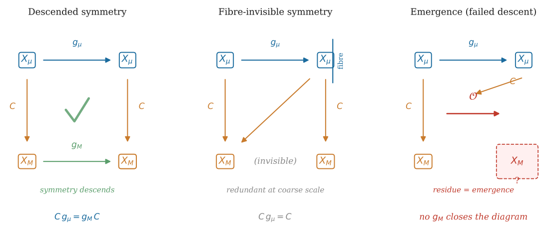

FIG. 7. **Transported symmetry, fibre-invisible symmetry, and failed closure (appendix figure).** *Left:* Fine symmetry satisfying $Cg_\mu = g_M C$; transported symmetry. *Centre:* Fine transformation satisfying $Cg_\mu = C$; fibre-invisible, becomes gauge redundancy. *Right:* No g_M closes the diagram; failed closure—residue is emergence.

4. Dual closure residue and conservation laws

The dual closure residue

$$\Delta_C{}^\dagger = G_\mu^\dagger C^* - C^* G_M^\dagger \tag{E3}$$

measures the failure of a macro integral of motion to lift to a true micro integral. For $G_M^\dagger I_M = 0$,

$$G_\mu^\dagger C^* I_M = \Delta_C{}^\dagger I_M. \tag{E4}$$

A macro conservation law $\partial_t \langle I_M \rangle = 0$ holds at the coarse level, but the lifted observable $C^* I_M$ is not conserved at the fine level unless $\Delta_C{}^\dagger I_M = 0$.

The set of integrals at scale s forms an algebra (closed under Poisson bracket or operator commutator). Changing scale changes which observables are available as integrals. Emergence is the appearance, disappearance, or incompatibility of conserved algebras under partition change: the closure residue of the transport of integrals of motion.

Integrable systems and MBL. In integrable systems, a large algebra of local integrals of motion survives arbitrary coarse-grainings in certain representations (the action variables of a classically integrable system). In many-body localised phases, approximate local integrals of motion are the effective conserved quantities at each scale of the disorder structure. The dual closure residue $\Delta_C{}^\dagger I_s$ measures the rate at which a local integral fails to remain conserved under vertical coarse-graining. Small $\|\Delta_C{}^\dagger I_s\|$ is the precise condition for approximate closure of a conserved quantity across scale.

Appendix F: Reconstruction, gauge, holonomy, and spectral residue

1. Affine reconstruction freedom

A coarse-graining $C : X_\mu \to X_M$ discards distinctions. Recovering them requires a reconstruction map $R : X_M \to X_\mu$ satisfying

$$CR = I_M. \tag{F1}$$

If R is one such map and $N : X_M \to X_\mu$ satisfies $CN = 0$ (N maps into the kernel of C), then $R' = R + N$ is also a valid reconstruction.

Proposition 6 (Affine reconstruction bundle)**.** *For a fixed linear coarse map C, the set of right inverses $\mathrm{Sec}(C) = \{R : CR = I_M\}$ is either empty or an affine space modelled on $\mathrm{Vert}(C) = \{N : CN = 0\}$.*

Proof. If $R, R' \in \mathrm{Sec}(C)$, then $C(R'-R) = 0$, so $R'-R \in \mathrm{Vert}(C)$. Conversely, $C(R+N) = CR+CN = I_M + 0 = I_M$ for any $N \in \mathrm{Vert}(C)$. □

The hidden object is not a catalogue of primitive values; it is the affine space of admissible liftings.

2. Gauge as self-similar reconstruction freedom

Two reconstructions R and $R' = R + N$ are *gauge-equivalent* if every coarse prediction of the descended law is the same for both: the difference N is physically redundant given the coarse law-content.

Definition 3 (Gauge)**.** *Gauge freedom is reconstruction freedom constrained by preservation of coarse law-content.*

Gauge invariance in electrodynamics, Yang-Mills theory, or gravity corresponds to the physical statement that coarse predictions are invariant under the relevant class of fine-level rearrangements. Principal-bundle structure requires additionally a group $G_C = \{U : CU = C\}$ acting freely and transitively on admissible sections. The affine bundle (Proposition 6) is the underlying structure; naming G_C gives the principal bundle.

3. Holonomy from imperfect reconstruction transport

To compare reconstruction choices at neighbouring representations r^i in a space of representations $\mathcal{R}$, a connection is needed. Let

$$D_i = \partial_i + A_i, \tag{F2}$$

where A_i encodes how admissible reconstruction choices rotate as the representation changes. The curvature is

$$\Omega_{ij} = [D_i, D_j] = \partial_i A_j - \partial_j A_i + [A_i, A_j]. \tag{F3}$$

Around a closed loop γ in representation space, the holonomy is

$$U_\gamma = \mathcal{P} \exp \oint_\gamma A. \tag{F4}$$

When a compact one-dimensional reversible sector is isolated, this reduces to $U_\gamma = e^{i\theta_\gamma}$: the phase accumulated by transporting a reconstruction choice around a closed loop.

Perfect self-similarity of reconstruction (exact gauge invariance) gives flat connection: $\Omega_{ij} = 0$ and $U_\gamma = 1$. Imperfect self-similarity gives nonzero curvature and nontrivial holonomy. The Berry phase and its non-Abelian extension [8–10] are recovered as special cases in which the space of representations is the space of adiabatic parameters and the reconstruction fibre is a Hilbert-space bundle.

Closure residue and curvature are distinct objects. Closure residue ($\Delta_C \neq 0$) is closure failure at the representational level. Curvature ($\Omega \neq 0$) appears only after a connection on reconstruction choices has been specified. Do not equate them.

4. Noncommuting erasures and spectral closure residue

Spectral closure residue can arise from classical coarse-graining without invoking quantum mechanics. Let the fine space be $\Omega = \{1, 2, 3\}$. Description A identifies $\{1, 2\}$ and leaves $\{3\}$ alone:

$$\Pi_A = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{F5}$$

Description B leaves $\{1\}$ alone and identifies $\{2, 3\}$:

$$\Pi_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 1/2 & 1/2 \end{pmatrix}. \tag{F6}$$

Both operations are classical averaging. Nevertheless,

$$[\Pi_A, \Pi_B] = \begin{pmatrix} 0 & -1/4 & 1/4 \\ 1/4 & 0 & -1/4 \\ -1/4 & 1/4 & 0 \end{pmatrix} \neq 0, \tag{F7}$$

with Frobenius norm 0.612. Incompatible lossy descriptions produce noncommuting projection-reconstruction operations when their erased distinctions are incompatible. The commutator norm is the finite spectral closure residue of the pair of descriptions.

When descriptions are spectrally incompatible, there is no single joint distribution over the primitive properties visible to both. The natural calculus is then states as positive normalised functionals over a noncommutative algebra—in finite dimensions resembling density-operator calculus with sequential update taking the projection form $\rho \mapsto \Pi_A \rho \Pi_A / \operatorname{Tr}(\rho \Pi_A)$. This paper does not derive the Born rule [11]; it locates the structural reason to expect noncommutative probability before additional axioms are supplied.

5. Infinitesimal residue density: full derivation

Following (19)–(22), expand the finite-step closure residue:

$$C_{s\to s+\delta s}\,G_s = G_s + \delta s\,L_s G_s + O(\delta s^2), \tag{F8}$$

$$\begin{aligned} G_{s+\delta s}\,C_{s\to s+\delta s} &= (G_s + \delta s\,\dot{G}_s)(I + \delta s\,L_s) + O(\delta s^2) \\ &= G_s + \delta s(\dot{G}_s + G_s L_s) + O(\delta s^2). \end{aligned} \tag{F9}$$

Subtracting:

$$\begin{aligned} \Delta_C{}^{s\to s+\delta s} &= \delta s(L_s G_s - G_s L_s - \dot{G}_s) + O(\delta s^2) \\ &= \delta s([L_s, G_s] - \dot{G}_s) + O(\delta s^2). \end{aligned} \tag{F10}$$

The residue density is $\omega_s = [L_s, G_s] - \dot{G}_s$. If $D_s = \partial_s - \mathrm{ad}(L_s)$ is the covariant scale-derivative, this is $\omega_s = -D_s G_s$. Exact differential closure is $D_s G_s = 0$: the generator is parallel-transported by the coarse-graining connection and leaves no residue.

6. Two-level spectral calculation

For $G_s = \frac{\Delta}{2}\sigma_z$ and $\omega_s = \kappa\sigma_x$, define the spectral connection (off-diagonal residue over spectral gap):

$$B_s = i\frac{\kappa}{\Delta}\,\sigma_y. \tag{F11}$$

Check: $[B_s, G_s] = [i(\kappa/\Delta)\sigma_y,\, (\Delta/2)\sigma_z] = -\kappa\sigma_x = -\omega_s$. ✓

With $\Pi_+ = (1 + \sigma_z)/2$, the transported projector is

$$\begin{aligned} \Pi_+(s + \delta s) &= \Pi_+(s) + \delta s[B_s, \Pi_+] + O(\delta s^2) \\ &= \Pi_+(s) - \delta s\,\frac{\kappa}{\Delta}\,\sigma_x + O(\delta s^2). \end{aligned} \tag{F12}$$

Therefore

$$[\Pi_+(s),\, \Pi_+(s + \delta s)] = -i\,\delta s\,\frac{\kappa}{\Delta}\,\sigma_y + O(\delta s^2), \tag{F13}$$

and

$$\lim_{\delta s\to 0} \frac{[\Pi_+(s),\, \Pi_+(s + \delta s)]}{\delta s} = -i\frac{\kappa}{\Delta}\,\sigma_y. \tag{F14}$$

The commutator density is finite; the raw commutator vanishes. The dimensionless spectral residue ratio is $\chi_s = \kappa/\Delta$. When $\chi_s \to 0$, spectral frames become asymptotically compatible (classical closure). When $\chi_s \to \chi^* \neq 0$, adjacent-scale descriptions differ infinitesimally but their spectral-frame rotation rate is finite and scale-stable: self-similar spectral residue.

Appendix G: Quantum-like fixed points

1. Self-similar residue dynamics

The residue density ω_s is itself subject to scale evolution. Let $\mathcal{R}_s$ be the space of residue densities at scale s and $\mathcal{S}_s : \mathcal{R}_s \to \mathcal{R}_{s+ds}$ the residue RG map.

Classical closure corresponds to the trivial fixed point $\omega^* = 0$: the residue decays under scale evolution and the system achieves classical autonomy in the limit. Quantum-like closure corresponds to a nontrivial fixed point:

$$\omega^* \neq 0, \qquad \mathcal{S}(\omega^*) = \omega^*. \tag{G1}$$

At this fixed point the failure of classical closure is itself governed by a self-similar law. The spectral noncommutativity produced by the off-diagonal part of ω^* is not a transient perturbation; it is a stable, scale-invariant feature of the algebra of descriptions.

More general self-similar residue structures include: anomalous scaling ($\omega_s \sim b^{-\Delta}\omega^*$), limit cycles ($\omega_{s+T} = \omega_s$), state-dependent fixed structures ($\omega_s = \omega_s[\rho]$), non-Abelian curvature ($F_{ab} \neq 0$), and nonassociative higher residue ($[A, B, C] \neq 0$).

2. Operational $\hbar$ as fixed-point closure residue

When spectral closure residue reaches the nontrivial fixed point, the off-diagonal residue density induces a stable commutator algebra on observable pairs. Define the *operational closure-residue scale* $\hbar_{\mathrm{op}}$: the coefficient converting stable spectral residue into an effective commutator algebra.

For conjugate finite descriptions Q and P, classical compatibility gives $[Q, P] = 0$. At the quantum fixed point, $[Q, P] = i\hbar_{\mathrm{op}}\,\Omega(Q, P)$ for a symplectic form Ω. Ordinary quantum mechanics is the special case in which $\hbar_{\mathrm{op}}$ becomes scalar, central, and universal:

$$\hbar_{\mathrm{op}}(s, \rho, C, R) \to \hbar\, I. \tag{G2}$$

A small loop in conjugate representation space returns with phase

$$\exp\left(-\frac{i}{\hbar}\,ab\right), \tag{G3}$$

where ab is the enclosed phase-space action area. $\hbar$ is the quantum of action because it is the fixed-point scale of irreducible representational residue. This identifies the structural role of $\hbar$; it does not determine the numerical value, which is set by physical content not representational calculus.

3. Classification of self-similar residue regimes

Table II organises the possible fixed-point and self-similar residue structures. Ordinary quantum mechanics is not the most general noncommutative theory of finite description; it is the unusually rigid case where spectral closure residue becomes scalar, central, universal, linear, associative, and Hilbert-representable.

TABLE II. Self-similar residue regimes. Each row corresponds to a distinct scale-stable structure of the spectral closure residue. Ordinary quantum mechanics is the rigid case in which the residue is scalar, central, universal, linear, associative, and Hilbert-representable.

Residue flow	Algebraic form	Regime
$\omega_s \to 0$	$[A, B] \to 0$	Classical closure
$\omega_s \to \omega^*$, $\hbar_{\rm op} \to \hbar I$	$[A, B] = i\hbar\,\Omega(A, B)$	Ordinary quantum mechanics
$\omega_s \to \omega^*$, $\hbar_{\rm op} \not\propto I$	$[A, B] = i\,\Theta(A, B)$	Noncanonical quantum-like theory
$\omega_s \sim b^{-\Delta}\omega^*$	$[A, B]_s = i\hbar_s\,\Omega(A, B)$	Scale-anomalous quantum theory
$\omega_{s+T} = \omega_s$	log-periodic commutator	Discrete-scale quantum-like theory
$\omega_s = \omega_s[\rho]$	$[A, B]_\rho = i\,\Omega_\rho(A, B)$	State-dependent / contextual theory
$F_{ab} \neq 0$	$[D_a, D_b] = F_{ab}$	Holonomic / gauge-like theory
$[A, B, C] \neq 0$	nonzero associator	Higher / nonassociative residue theory

4. Nonclaims

The framework:

- Identifies the structural setting for quantum-like behaviour as the nontrivial fixed point of self-similar spectral closure residue.
- Shows why noncommutative probability is the natural calculus when finite scale-fixings are spectrally incompatible.
- Identifies $\hbar$ as the universal scalar fixed-point scale of spectral closure residue.
- Identifies no-go theorems (Bell [12], Kochen-Specker [13, 14]) as obstructions to a global common spectral frame—the impossibility of making all reconstruction choices jointly flat.

The framework does not:

- Derive the Born rule [11].
- Reconstruct Hilbert space from first principles.
- Determine the numerical value of $\hbar$.
- Prove that all non-neutral scale fixing is quantum; more general self-similar residue regimes are possible (Table II).
- Claim that all commutator structures reduce to Hilbert-space quantum mechanics; standard quantum mechanics is the rigid fixed point, not the most general possible noncommutative theory.

The points where additional axioms would enter are: the admissible structure group on reconstruction choices, the algebra of observables, and the regularity of the residue flow. The present contribution is to have identified the structural reason to expect projector-noncommutativity, not to have derived it from scratch. Gleason's theorem [11] and related results show what additional assumptions can do once the appropriate projector structure is in place.

Appendix H: Scaling laws from closure-residue decay

This appendix provides technical details for the closure-residue interpretation of neural scaling laws and interpretability phase transitions.

1. Cross-entropy as KL residue

The training of a neural network can be viewed formally as a finite-description closure problem. Given a data distribution $p_{\rm data}$ and a parametrised model family p_θ, the standard cross-entropy loss is

$$\mathcal{L}(\theta) = -\mathbb{E}_{p_{\rm data}} \log p_\theta. \tag{H1}$$

This decomposes via standard identities into the irreducible entropy of the data generating process and the Kullback-Leibler divergence between the data and the model,

$$\mathcal{L}(\theta) = H(p_{\rm data}) + D_{\rm KL}(p_{\rm data} \| p_\theta). \tag{H2}$$

The $D_{\rm KL}$ term is the scalar representational residue: the excess statistical content of the data law that cannot be absorbed into the finite representation p_θ. The irreducible entropy $H(p_{\rm data})$ is the null statistical content, fixed by the data source independently of model choice.

2. Residue-mode decomposition

Consider a representation-scale parameter s. The closure residue can be decomposed into an eigenbasis of modes,

$$\Delta_s = \sum_k a_k(s)\phi_k. \tag{H3}$$

Linearising the residue renormalisation group flow near a closure fixed point yields

$$\partial_s a_k = -\lambda_k a_k, \tag{H4}$$

where λ_k are the corresponding eigenvalues. The solution implies exponential decay in the scale parameter,

$$a_k(s) = a_k(0)e^{-\lambda_k s}. \tag{H5}$$

By defining the scale parameter logarithmically with respect to a physical model resource, $s = \log N$ (where N is parameter count or capacity), we obtain

$$a_k(N) = a_k(0)N^{-\lambda_k}. \tag{H6}$$

If the total loss is dominated by the leading (slowest-decaying) residue mode, the expected scaling law emerges [15]:

$$\mathcal{L}(N) - \mathcal{L}_\infty \sim AN^{-\alpha}. \tag{H7}$$

3. Multi-axis scaling

In practice, model capacity, dataset size, and compute define a coupled system of representation scales: $s_N = \log N$, $s_D = \log D$, and $s_C = \log C$. The residue flow then depends on multiple axes,

$$\mathcal{L}(N, D) - \mathcal{L}_\infty \approx AN^{-\alpha} + BD^{-\beta} + E_{\text{int}}(N, D). \tag{H8}$$

Here, the $AN^{-\alpha}$ term is the capacity closure residue, the $BD^{-\beta}$ term is the sampling (or null-law) residue, and E_{int} represents the interaction residue stemming from imperfect factorisation between the capacity and data scales.

4. Compute-optimal balance

Following the derivation of compute-optimal scaling [16], we apply a simple compute constraint $C \sim ND$. We wish to minimise the total residue

$$AN^{-\alpha} + BD^{-\beta} \tag{H9}$$

subject to $ND = C$. Substituting $D = C/N$, we obtain

$$\mathcal{L}(N) \approx AN^{-\alpha} + BC^{-\beta}N^{\beta}. \tag{H10}$$

Taking the derivative with respect to N and setting to zero yields the optimal balance,

$$\alpha AN^{-\alpha} \sim \beta BD^{-\beta}. \tag{H11}$$

This provides a closure-residue interpretation for compute-optimal scaling: the allocation of resources is optimal when the marginal reduction of capacity residue exactly balances the marginal reduction of sampling residue. Neither should dominate.

5. Circuit formation as residue-class transition

The closure calculus predicts that the emergence of discrete structural phenomena—such as induction heads, feature splitting, or superposition [17, 18]—corresponds to a transition in the class of the residue.

Defining a smooth scaling residual,

$$R(N, D) = \mathcal{L}_{\text{obs}}(N, D) - \mathcal{L}_{\text{fit}}(N, D), \tag{H12}$$

provides a testable empirical programme: mechanistic circuit formation should occur near structured features of $R(N, D)$. Specifically, local extrema, knees, exponent changes, or curvature changes in the log-log loss curve should mark the points where residue transitions from incoherent/nonlocal forms into coherent visible structure.

6. Relation to existing literature

The smooth power-law scaling of loss with model and data size was established empirically by Kaplan *et al.* [15] and subsequently refined by compute-optimal analyses [16]. These empirical laws have been studied in terms of data-limited versus parameter-limited regimes, but the present framework offers a structural account: each scaling axis corresponds to an independent residue-decay mode, and compute-optimal balance is the condition that no single mode dominates.

Mechanistic interpretability has documented a range of discrete structural transitions in trained networks—induction heads [19], superposition [18], and feature splitting—that appear as sharp qualitative changes rather than smooth variations. From the closure-residue perspective, these are residue-class transitions: the dominant residue mode changes algebraic type (from nonlocal/incoherent to coherent/transition), producing the observed discontinuous character. The present framework predicts that such transitions should be visible as structured features in the residual $R(N, D)$ defined above, and thus provides a quantitative bridge between the scaling-laws literature and the mechanistic-interpretability literature.

7. Empirical prediction

The central empirical prediction of the closure-residue framework is:

> *Structured deviations from smooth scaling residuals $R(N, D)$ should correlate with mechanistic reorganisation events—circuit formation, feature splitting, induction-head emergence, and superposition reorganisations—rather than occurring randomly along the scaling curve.*

Smooth scaling is the self-similar decay of residue modes in the absence of residue-class transitions. Deviations from smooth scaling are not noise by default; they mark points where the algebraic class of the dominant residue changes. This prediction is falsifiable: if mechanistic transitions occur at randomly distributed points along the scaling curve, the framework's identification of residue-class transitions with observable scaling features would be refuted.

Appendix I: Compression costs and mass-like parameters

The cost-function structure of K_s identifies where mass-like parameters enter the closure calculus. The structural mapping is:

$$K_s = \sum_a g_a A_a \tag{I1}$$

$$A_{\text{comp}} \xrightarrow{\text{enforced}} g_{\text{comp}} \in K_s \xrightarrow{\text{if closed}} m_{\text{eff}}, \tag{I2}$$

where A_{comp} is the retained compression variable, g_{comp} its conjugate cost, and m_{eff} the resulting mass-like coefficient.

A localised excitation is a compressed pattern of lower-scale structure that remains identifiable after coarse-graining. If the finite description retains a variable A_{comp} measuring the persistence of such a compression, the conjugate coefficient g_{comp} entering K_s is the cost of maintaining it. When this cost:

1. closes under the coarse-graining (the cost survives as a well-defined feature of the coarse description),

2. is a scalar (not a matrix or tensor in the spectral frame),

3. is transportable (the same cost appears consistently across representation changes),

it has the structural role of a mass-like parameter. In this limited sense, mass is compression after closure: not the compression pattern itself, but the cost of maintaining a self-closing compression against the null law.

To obtain the familiar local form of a mass term would require the following additional steps:

1. Derive the cost-of-complexity function explicitly from the null-law recursion.

2. Obtain its differential form by expanding the residue triage in δs.

3. Lift the construction to path/action space (Appendix B).

4. Take the relevant small-deformation (gradient-expansion) limit.

These steps belong to a more developed treatment of the path-space version of the theory. The present contribution is more limited:

> The framework identifies where mass-like coefficients enter the closure calculus: as closed scalar costs of retained compression.

This identification is independent of the specific physical content (field type, spacetime symmetry, gauge structure) that would determine the numerical value of the mass. Those inputs enter at the step of specifying the observable A_{comp} and the admissible class $\mathcal{N}_s$, not at the level of the general closure calculus.

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